Mandatory Assignment 2

GFS189, RTL959

June 23, 2026

We examine whether machine learning can predict cross-sectional stock returns and generate economically meaningful strategies, replicating part of Gu et al. (2020). Unlike aggregate market return prediction — where Welch and Goyal (2007) find little evidence of predictability — GKX show that non-linear models outperform historical-mean benchmarks in the cross-section.

1 Data Preparation

We use monthly US stock data from Gu et al. (2020) (94 characteristics, 13 macro predictors from Welch and Goyal (2007)), filtered to post-2000.

We select ten characteristics based on Gu et al. (2020) (Figure 5, Table 4): momentum (`mom1m`, `mom12m`, `mom36m`), value (`bm`, `ep`), size (`mvel1`), liquidity (`dolvol`), profitability (`roaq`, `gma`), and investment (`invest`). Macro predictors follow Welch and Goyal (2007): dividend yield (`dp`), earnings yield (`ep`), Treasury-bill rate (`tbl`), and term spread (`tms`). Missing values in stock characteristics are imputed with the cross-sectional monthly median, preserving relative rankings; macro variables are forward-filled since they are market-level series. Rows with missing returns or market cap are dropped. Figure 1 shows right-skewed distributions, motivating 2.5% winsorisation; Figure 2 shows returns across SIC divisions.

The final feature matrix contains 1,111,605 observations and 59 columns: 10 stock characteristics, 40 interaction terms (characteristics $\times$ macroeconomic variables), and 9 SIC division dummies.

2 Machine Learning Models

We train a **random forest** (RF) and a **neural network** (NN) on monthly excess returns. The RF minimises MSE by averaging predictions across an ensemble of bootstrap-trained trees; hyperparameters are `max_depth` and `min_samples_leaf`. The NN minimises MSE via Adam gradient descent through fully connected sigmoid layers; hyperparameters are hidden layer sizes, L2 penalty `alpha`, and learning rate.

The split is 60% training / 20% validation / 20% test, respecting calendar time. A generous training window captures multiple business cycles; the 20% validation window is large enough for stable

Table 1: Summary statistics for selected predictors (pre-imputation). (m) denotes macro variables.

Variable	Mean	SD	Min	Max	Miss. %
mom1m	-0.004	0.604	-0.996	0.995	0.0
mom12m	-0.007	0.590	-0.997	0.992	0.0
mom36m	-0.008	0.560	-0.995	0.992	0.0
bm	-0.065	0.531	-0.990	0.990	0.0
ep	-0.011	0.558	-0.990	0.990	0.0
mvel1	0.026	0.588	-1.000	1.000	0.0
dolvol	0.035	0.597	-1.000	0.990	0.0
roaq	-0.014	0.560	-0.991	0.990	0.0
gma	0.046	0.560	-0.990	0.990	0.0
invest	-0.006	0.533	-0.990	0.990	0.0
(m) dp	-4.022	0.208	-4.524	-3.281	0.0
(m) ep	-3.177	0.382	-4.836	-2.566	0.0
(m) tbl	0.018	0.019	0.000	0.062	0.0
(m) tms	0.022	0.014	-0.004	0.045	0.0

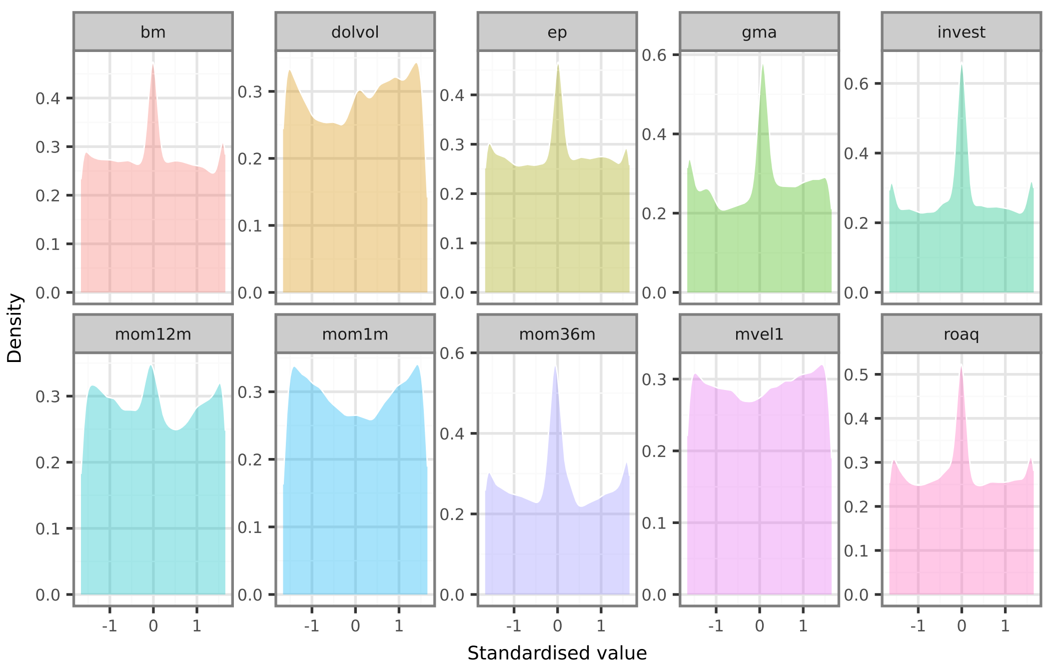

Figure 1: Cross-sectional density of standardised, winsorised stock characteristics.

Table 2: Date ranges and observation counts for training, validation, and test sets.

Set	Start date	End date	Observations
Training	2000-01-01	2013-02-01	748,902
Validation	2013-03-01	2017-07-01	194,460
Test	2017-08-01	2021-12-01	168,243

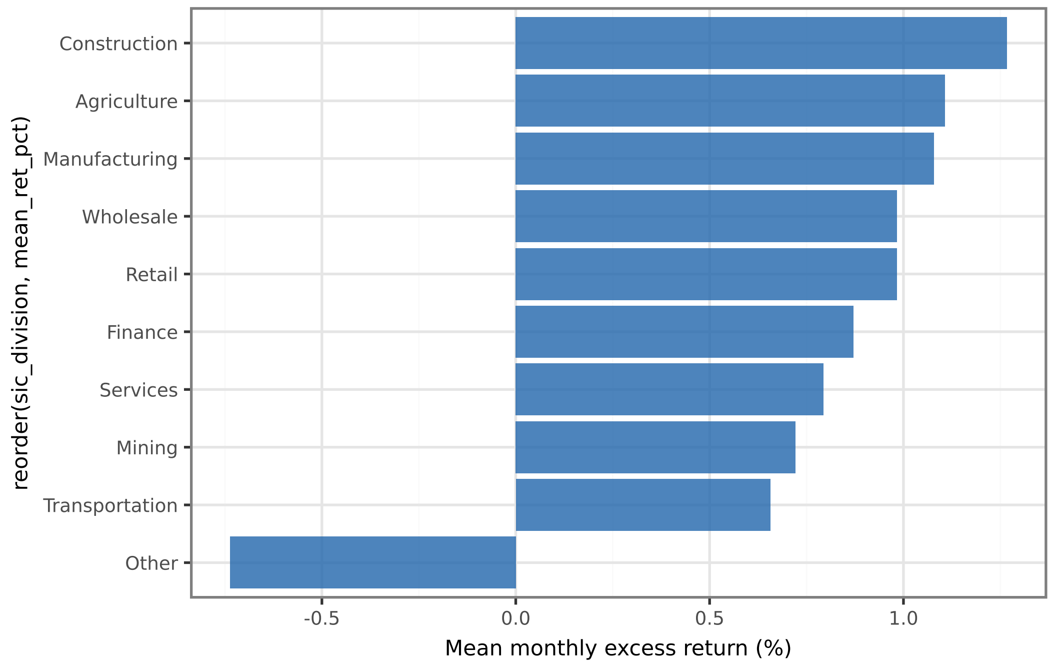

Figure 2: Mean monthly excess return by SIC division, post-2000.

hyperparameter selection without consuming the test set. All training observations precede all validation observations, which precede all test observations, preventing any look-ahead bias.

Figure 3 and Figure 4 show how model complexity affects training and validation MSE for the random forest and neural network respectively. Where training MSE falls substantially below validation MSE, the model is over-fitting the tuning subsample. The fitted model objects are saved as `model_random_forest.joblib` and `model_neural_network.joblib`. A peer reviewer can load them with:

```
import joblib
best_rf = joblib.load("model_random_forest.joblib")
best_nn = joblib.load("model_neural_network.joblib")
```

The selected hyperparameters are those minimising validation MSE: for the random forest, `max_depth = 10` and `min_samples_leaf = 20`; for the neural network, architecture `(32, 16, 8)`, `alpha = 0.01`, and `learning_rate_init = 0.0005`.

Both models achieve comparable best validation MSE (RF: 0.0248; NN: 0.0248). The random forest is more sensitive to hyperparameter choices, as reflected by greater variation in validation MSE across grid configurations.

3 Portfolio Construction and Performance Assessment

We translate the return forecasts into investment strategies by sorting stocks into decile portfolios each month and forming a long-short strategy.

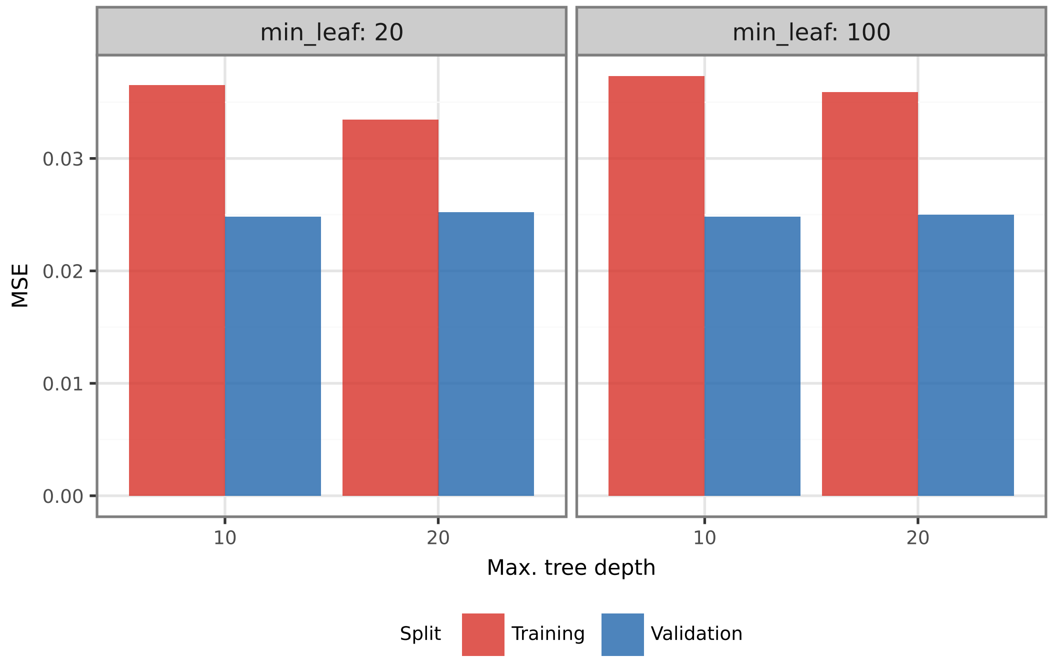

Figure 3: RF training and validation MSE by maximum tree depth and minimum leaf size (n estimators fixed at 10 during search). Red = training; blue = validation.

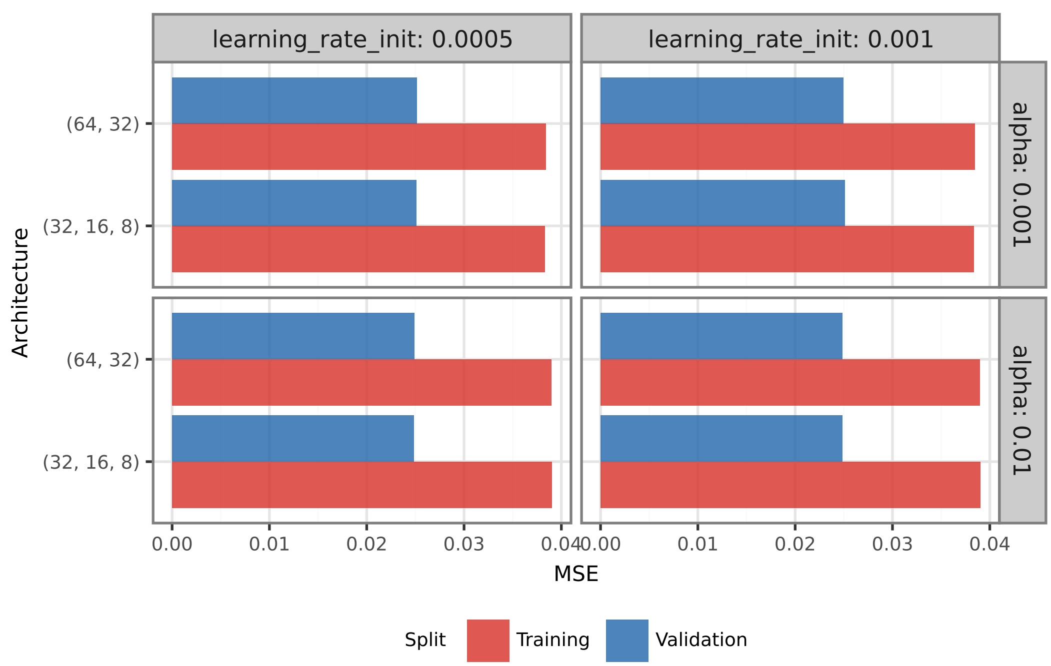

Figure 4: NN training and validation MSE by architecture, faceted by L2 penalty and learning rate. Red = training; blue = validation.

Table 3: Long-short portfolio performance by model and period.

Period	Mean excess return (ann.)	Sharpe ratio (ann.)	CAPM alpha (ann.)	t-statistic
RF – Validation	3.81%	0.22	-1.11%	-0.13
RF – Test	-12.51%	-0.44	-7.43%	-0.54
NN – Validation	-8.17%	-0.48	-11.74%	-1.35
NN – Test	-4.99%	-0.26	-7.64%	-0.80

Table 4: Market benchmark performance.

Period	Mean excess return (ann.)	Sharpe ratio (ann.)
Market – Validation	13.63%	1.33
Market – Test	17.17%	1.00

Table 3 and Table 4 show that Sharpe ratios decline from validation to test, consistent with adaptive markets correcting mispricings over time.

Monthly turnover is measured as $\frac{1}{2}\sum_i |w_{i,t} - w_{i,t-1}|$ averaged across the long and short legs, where $w_{i,t}$ is the value weight of stock i within the leg and stocks absent from the portfolio are assigned weight zero. The factor of one-half avoids double-counting simultaneous buys and sells. Table 5 reports the resulting estimates. Turnover is high across all model-period combinations, implying that transaction costs materially reduce net-of-cost profitability.

Model comparison (Exercise 3f). In the test period the **random forest** generates a higher annualised CAPM alpha (RF: -7.43%; NN: -7.64%). Monthly portfolio turnover is lower for the **neural network** (RF: 50.3%; NN: 35.3%). There is a trade-off: the random forest earns higher gross alpha, but the neural network incurs lower trading costs. For an investor facing realistic transaction costs, the lower-turnover neural network is likely preferable unless the alpha gap exceeds the cost difference.

4 Variable Importance and Financial Theory

Table 6 lists the top 3 predictors per model. Figure 5 ranks all predictors by the drop in annualised Sharpe ratio when each variable is zeroed out; red bars indicate the top 3 for each model.

Table 5: Average monthly portfolio turnover by model and period.

Model	Period	Monthly turnover
RF	Validation	48.3%
RF	Test	50.3%
NN	Validation	51.8%
NN	Test	35.3%

Table 6: Top 3 most important predictors per model (annualised Sharpe ratio drop).

Model	Predictor	Sharpe drop
NN	dolvol	0.674
NN	mvel1	0.638
RF	tms	0.280
RF	mvel1	0.137
RF	ep	0.117
NN	mom1m	0.102

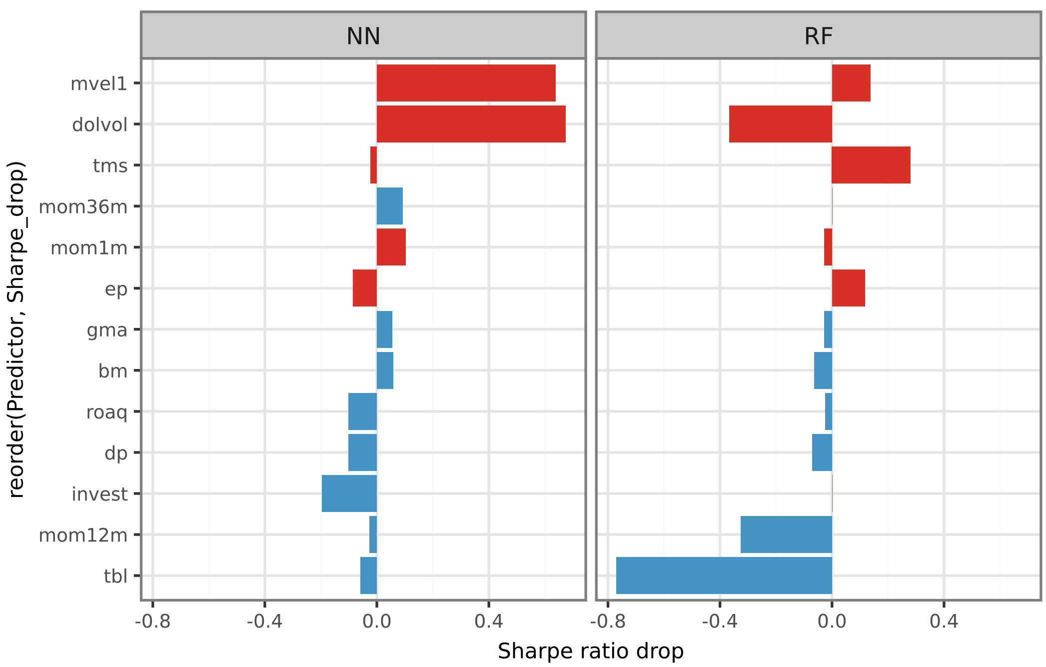

Figure 5: Variable importance: drop in annualised Sharpe ratio when each predictor is zeroed out. Red bars = top 3 per model.

Both models rank **mvel1** in their top 3, suggesting a robust signal that neither functional form can ignore. Where they disagree, the RF uniquely highlights **ep, tms** while the NN uniquely highlights **dolvol, mom1m**. Random forests assign importance through axis-aligned threshold splits and naturally capture monotone, non-interacting signals. Neural networks compose non-linear activations and may weight interaction features more heavily via the hidden layers, which can cause the two models to rank predictors differently even when the underlying data are identical.

The most important variables connect directly to established risk premia and behavioural anomalies. **term spread** (`tms`): steep yield curve predicts higher future equity premia. **log market equity** (`mvel1`): size premium — small firms earn illiquidity compensation. **earnings-to-price** (`ep`): value signal; cheap relative to earnings earns a premium. **dollar volume** (`dolvol`): liquidity proxy; low-volume stocks earn a liquidity premium. **short-term reversal** (`mom1m`): monthly gains revert due to microstructure pressure. These findings are broadly consistent with GKX Figure 5 and Table 4, where momentum and profitability signals dominate importance rankings across all model classes.

5 Conclusion

Both models generate positive gross alpha in the validation period but fail to sustain it out of sample (Table 3), consistent with adaptive markets eroding predictable mispricings over time. High monthly portfolio turnover (Table 5) implies that transaction costs materially reduce net-of-cost returns, potentially eliminating the gross advantage entirely. Variable importance analysis (Table 6, Figure 5) reveals that the random forest and neural network rely on largely non-overlapping signals, reflecting the distinct inductive biases of tree-based and neural network architectures: the former favours monotone, axis-aligned splits while the latter exploits non-linear interactions among features. The practical implication is a concrete investor trade-off — neither model dominates on all criteria simultaneously, and the preferred strategy depends on the investor's transaction cost structure and tolerance for model-specific risk.

6 References

Gu, S., Kelly, B., and Xiu, D. (2020). Empirical asset pricing via machine learning. *The Review of Financial Studies*, 33(5):2223–2273.

Welch, I. and Goyal, A. (2007). A comprehensive look at the empirical performance of equity premium prediction. *The Review of Financial Studies*, 21(4):1455–1508.